

Civics Group	Index Number	Name (use BLOCK LETTERS)
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H2



ST. ANDREW'S JUNIOR COLLEGE
2018 JC2 Prelim

H2 BIOLOGY

9744/03

Paper 3

Monday

17th September 2018

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A (Structured Questions)

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

Section B (Essay Question)

Answer **one** essay question.

Write your answers in the spaces provided on the question paper.

All working for numerical answers must be shown.

For Examiners' Use	
Section A	
1	/35
2	/7
3	/8
Section B	
4 or 5	/25
Total	/75

This document consists of **20** printed pages.

[Turn over

Section A

*Answer **all** questions.*

QUESTION 1

- (a)** Potential cancer cells experience a series of alterations during oncogenic transformation that confers them new features. These features are both genetic and phenotypic in nature. One prominent feature is the reprogramming of their energy metabolism. This phenomenon has been termed 'The Warburg Effect'.

The Warburg Effect describes the observation that cancer cells exhibit an enhanced glycolysis without undergoing oxidative phosphorylation, even when oxygen is abundant. This leads to another prominent feature of cancer – the acidification of the extracellular environment of tumours due to the secretion of an overproduced molecule by cancer cells as a result of enhanced glycolysis.

Since the observation, many journal reports have surfaced to provide explanations to the Warburg Effect. Some of these include various mutations, upregulation or downregulation of expression of various genes involved in respiration, for example, those that code for the enzyme ATP synthase.

- (i)** Outline the 'series of alterations' that could occur in the potential cancer cells.

[illegible]

- (ii)** Explain how an enhanced glycolysis can lead to the acidification of the tumours' extracellular environment.

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The fact that most cancer cells exhibit the Warburg Effect has been used clinically as a basis for cancer diagnosis in patients. This method of diagnosis is known as PET scanning as outlined below:

- Patient is first injected with ^{18}F -fluorodeoxyglucose, a radioactive glucose analogue
- ^{18}F -fluorodeoxyglucose can be taken up by respiring cells
- In the cytoplasm, ^{18}F -fluorodeoxyglucose can be phosphorylated by hexokinase but cannot be metabolised further in glycolysis
- Presence of radioactivity is determined in the various sites of the body

- (iii)** Predict the observation in PET scanning on cancer cells compared to non-cancer cells.

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- (iv)** Explain the prediction made in **(iii)**.

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- (v)** Suggest why cancer cells require more glucose than non-cancer cells to meet energy demands.

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- (b) Central to the energy metabolism of aerobically respiring cells is the enzyme ATP synthase. It consists of multiple subunits, coded for by several genes.

The enzyme is located on the inner mitochondrial membrane and catalyses the reversible reaction of ATP synthesis from ADP in intact mitochondria. Here, the **proton motive force** is required to drive the ATP synthesis.

- (i) Explain how the structure, including protein components, of the inner mitochondrial membrane is significant in driving the reaction of ATP synthase towards ATP synthesis during aerobic respiration.

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In a cell-free or mitochondrial-free system, the same enzyme can also catalyse ATP hydrolysis when ATP is present. This reaction can be exploited for the investigation of ATP synthase activity.

To examine the enzymatic activity of ATP synthase, the indirect method of measuring the relative NADH concentration in the solution can be utilised as shown below in **Fig. 1.1**.

ATP produced when phosphoenolpyruvate is converted to pyruvate by pyruvate kinase, is acted upon by ATP synthase to form ADP.

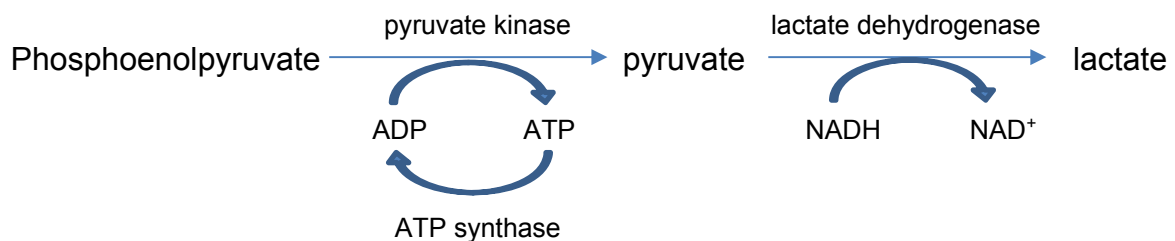


Fig. 1.1

The relative concentration of NADH can be tracked with a spectrophotometer as NADH absorbs light at 340nm while NAD⁺ and other molecules in the system do not.

- (ii) In examining the activity of ATP synthase using the reactions in **Fig. 1.1**, explain why phosphoenolpyruvate, ATP and NADH need to be in high concentrations.

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The reactions in **Fig. 1.1** can also be used to determine the effect of a particular drug on ATP synthase. Curcumin, a phytochemical isolated from the rhizome of turmeric, has been shown to affect the activity of ATP synthase. However, its effect varies according to the source of ATP synthase.

Fig. 1.2 and **Fig. 1.3** show the effect of curcumin on ATP synthase molecules isolated from the liver mitochondria and brain mitochondria of the same rat respectively.

% control refers to ATP synthase activity with reference to activity at 0 μM curcumin.

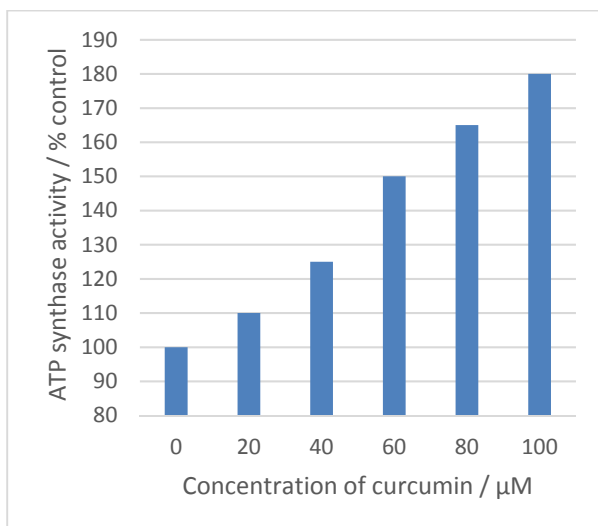


Fig. 1.2: Effect of curcumin on ATP synthase isolated from rat liver mitochondria

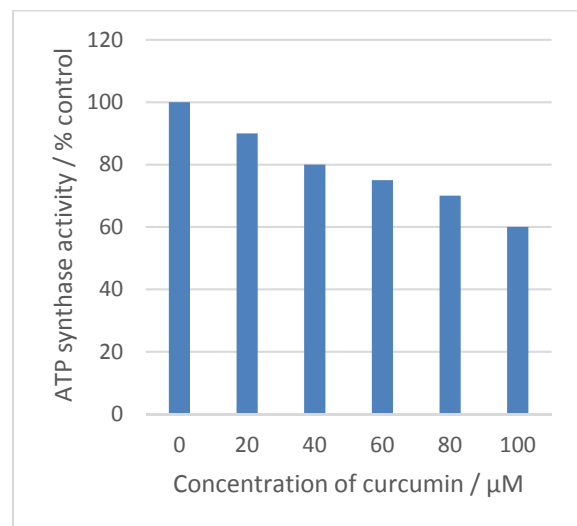


Fig. 1.3: Effect of curcumin on ATP synthase isolated from rat brain mitochondria

- (iii) With reference to **Fig. 1.2**, describe the effect of curcumin on rat liver mitochondrial ATP synthase.

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- (iv)** Suggest one process that can occur within cells, and explain how it can lead to the differential effects of curcumin on ATP synthase molecules isolated from liver and brain mitochondria of the same rat.

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In many prokaryotes, the multi-subunit ATP synthase expression is under the control of a single operon.

- (v)** Suggest how the ATP synthase operon is organised.

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- (c) ATP synthase deficiency is a disorder in which individuals are deficient in the enzyme, leading to a wide variety of signs and symptoms affecting many organs and systems of the body. The disorder can range from being mild to life-threatening.

Different mutations in different nuclear genes can result in the deficiency. These can generally be grouped into 2 categories. The first being mutations at gene loci coding for an ATP synthase subunit, for example, *ATP5E*.

The second being mutations at gene loci coding for a protein that is required for the proper assembly of the ATP synthase subunits to form the functional enzyme, for example, *TMEM70*.

- (i) Justify the claim that the severity of the disorder in an individual depends on his/her genetic makeup.

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ATP synthase deficiency cases have been reported frequently in the Roma population, an ethnic group living mostly in Europe. Various journal articles have collectively reported the following:

- *TMEM70* gene defect was identified as a novel cause of autosomal recessive ATP synthase deficiency in 2009
- Most of the patients with *TMEM70* gene mutations share a common Roma descent
- In all genotyped cases for the patients of Roma origin, an adenine to guanine substitution was found, located in the splicing site of intron 2, which leads to aberrant splicing and thus preventing synthesis of the functional protein
- If the disorder is present as a result of this substitution mutation, a fatal outcome or life-threatening symptoms are expected

References:

- 1) Anne K. Braczynski, Stefan Vlaho, Klaus Müller, et al. (2015). ATP Synthase Deficiency due to *TMEM70* Mutation Leads to Ultrastructural Mitochondrial Degeneration and Is Amenable to Treatment. *BioMed Research International*, vol. 2015, Article ID 462592, 10 pages. <https://doi.org/10.1155/2015/462592>.
- 2) Josef Houštěk et al. (2009). *TMEM70* protein — A novel ancillary factor of mammalian ATP synthase. *Biochimica et Biophysica Acta – Bioenergetics*. 1787(5): 529-532.
- 3) Spiegel R et al. (2011). *TMEM70* mutations are a common cause of nuclear encoded ATP synthase assembly defect: further delineation of a new syndrome. *J Med Genet*. 48(3):177-82.

- (ii) It can thus be deduced that a particular mutation is associated with people of Roma origin that can result in ATP synthase deficiency.

With reference to the information given, suggest how the mutation could have been preserved in present day people of Roma descent despite leading to fatal outcomes.

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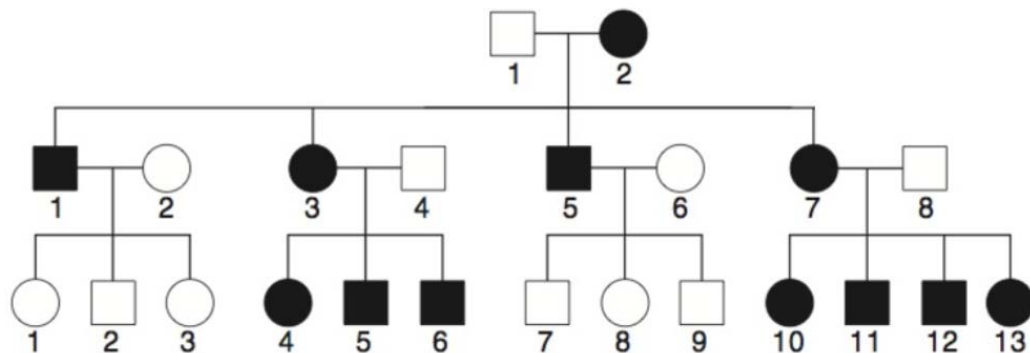
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ATP synthase deficiency can also result from mutations in the mitochondrial genes, for example, *ATP6*. The *ATP6* gene codes for a subunit of ATP synthase.

Fig. 1.4 shows a pedigree chart where individual 2, who is female, carries a mutation in the *ATP6* gene and thus suffer from the disorder. Individual 1 does not carry any mutations in the *ATP6* gene. Both individuals 1 and 2 do not carry any other mutations that can result in the disorder.



Legend:

- male without disorder
- male with disorder
- female without disorder
- female with disorder

Fig. 1.4

(iii) With reference to **Fig. 1.4**, explain how the disorder is inherited.

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[Total: 35]

QUESTION 2

Climate change, in the form of global warming, is expected to have an impact on various organisms. This impact however, varies geographically, particularly for insects. **Fig. 2.1** and **Fig. 2.2** below show the fitness curves of representative insects from a temperate and a tropical location respectively.

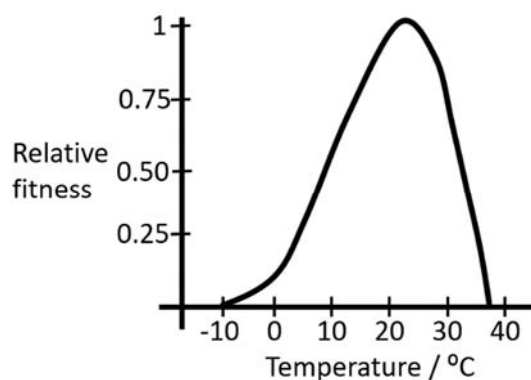


Fig. 2.1: Fitness curve of representative insect from temperate location

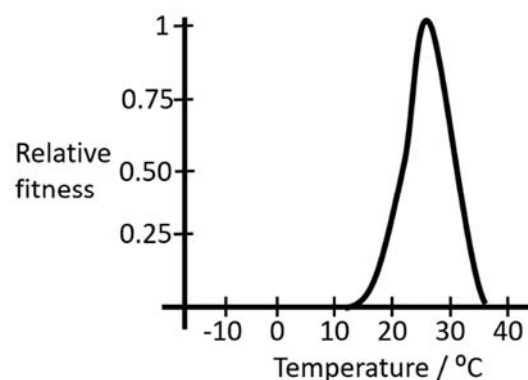


Fig. 2.2: Fitness curve of representative insect from tropical location

The mean annual temperature of the temperate location is 11 °C, with a typical temperature range from 1 to 20 °C. On the other hand, the mean annual temperature of the tropical location is 27 °C, with a typical temperature range from 21 to 31 °C.

(a) Suggest what is meant by the term relative fitness in **Fig. 2.1** and **Fig. 2.2**.

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- (b) With reference to **Fig. 2.1** and **Fig. 2.2** and the information given, predict and explain which insects, from the temperate or tropical location, would face a greater extinction risk as a result of global warming, assuming the warming elevates temperatures equally at both locations.

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- (c) Suggest one strategy that the insects in (b) can employ to reduce the impact of global warming on themselves.

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[Total: 7]

QUESTION 3

Proper activation of CD4 T cells is essential for an effective humoral immune response. This activation requires the involvement of antigen presenting cells, such as dendritic cells, that provide signals for the activation.

One such important signal is the presence of co-stimulatory molecules, such as CD80 or CD86, on the dendritic cell surface. These molecules are expressed and upregulated in dendritic cells via the Toll-like receptor signalling pathway. Toll-like receptors belong to a group of receptors known as Pattern Recognition Receptors that recognise typical surface molecules of pathogens as ligands.

CD4 T cell activation can occur via the natural route during infection or artificially via vaccination. Vaccines, when formulated effectively, could provide long term protection against a particular disease. Modern day vaccines that utilise only purified antigens generally do not evoke a strong immune response as compared to older style vaccines utilising live or killed whole bacteria.

- (a)** Outline how activation of CD4 T cells is important for an effective humoral immune response.

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- (b)** Explain how vaccines can provide a long term protection against a disease.

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- (c) For vaccines utilising purified antigens to be as effective as older style vaccines utilising live or killed whole bacteria, adjuvants need to be added to these vaccines.

Based on the information given, suggest what can be used as an adjuvant and explain how it contributes to the vaccine's effectiveness.

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[Total: 8]

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Section B

*Answer **one** question only.*

Write your answers on the lined paper provided at the end of this Question Paper.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)**, as indicated in the question.

QUESTION 4

- (a)** Cellulose is the most abundant biopolymer on earth. It forms a significant proportion of the dry mass of plants.

Describe the Calvin cycle and explain the origin of the carbon atoms in cellulose, taking into consideration the number of molecules involved. [12]

- (b)** In a particular plant species, which could produce offsprings in the hundreds, its height and flower colour are each controlled by a single gene locus. The two gene loci are located on different chromosomes. It is known that the alleles for tallness (T) and red flower (R) are dominant to the alleles for shortness (t) and white flower (r) respectively.

Using one molecular method and one genetic cross method, explain how the genotype of a tall, red-flowered plant of this species can be determined. [13]

[Total: 25]

QUESTION 5

- (a)** The Centers for Disease Control and Prevention in the United States of America recommends an annual influenza vaccination for everyone 6 months and older with rare exceptions, for example, those with life-threatening allergies to the vaccine. Each annual vaccine is different and contains three influenza strains.

Explain the need for a person to receive influenza vaccinations annually for protection against the disease and suggest why a person may still get the disease even after receiving his/her annual dose of vaccine. [12]

- (b)** Free nucleotides are present in various locations in a eukaryotic cell. They can exist both in the more commonly known non-cyclic form or in the cyclic form.

Describe the general structure of a non-cyclic form of nucleotide and with named examples, outline the roles of free nucleotides, cyclic or non-cyclic, at particular locations in a human cell. [13]

[Total: 25]

